A Two-Dimensional Smoothed Particle Hydrodynamics Solver

Author Name

November 4, 2026

Abstract

This paper documents the design and evaluation of a small two-dimensional smoothed particle hydrodynamics (SPH) solver. The fluid is modelled as weakly compressible with a Tait equation of state, the time step adapts to the acoustic, force, and viscous stability limits, and walls are represented by fixed particles with pressures extrapolated from the fluid. The solver is evaluated on a hydrostatic column and a dam break, and parameter studies quantify how the speed of sound, the viscosity, and the ratio of smoothing length to particle spacing affect compressibility, pressure noise, cost, and particle ordering.

Contents

1 Introduction **2**

2 Research Questions **2**

3 Methodology **2**
3.1 SPH Approximation . . . 2
3.2 Density and Equation of State . . . 2
3.3 Momentum Equation . . . 3
3.4 Boundary Conditions . . . 3
3.5 Time Integration . . . 4
3.6 Neighbour Search . . . 4
3.7 Software Structure . . . 4
3.8 Experimental Setup . . . 4

4 Results **4**
4.1 Correctness Tests . . . 5
4.2 Hydrostatic Column . . . 5
4.3 Dam Break . . . 6
4.4 Speed of Sound . . . 6
4.5 Viscosity . . . 7
4.6 Kernel Support and Pairing Instability . . . 7
4.7 Density Fields . . . 7
4.8 Discussion . . . 7

5 Conclusion **8**

1 Introduction

Particle-based methods are useful for simulating fluids with free surfaces and changing topology. Smoothed particle hydrodynamics represents a continuum by a set of moving particles and approximates field quantities using a smoothing kernel [5, 7].

The goal of this project is to implement and inspect a compact two-dimensional SPH solver in C++17. The solver treats the fluid as weakly compressible [8, 3]: pressure follows a stiff equation of state, the time step adapts to the acoustic stability limit, and solid walls are made of fixed particles whose pressure is extrapolated from the fluid [1]. An interactive SFML renderer displays the simulation, and a headless experiment runner produces the data for this paper. This paper describes the numerical model and the software structure, and evaluates the solver on a hydrostatic column and a dam break.

2 Research Questions

The evaluation focuses on three questions:

1. How accurately does the solver reproduce a known solution (hydrostatic pressure) and a classical experiment (dam break, [6]), and how large is the density error?
2. How do the numerical parameters (speed of sound, viscosity, and the ratio of smoothing length to particle spacing) affect compressibility, pressure noise, cost, and the flow itself?
3. Which artefacts limit the method, and where do they come from?

3 Methodology

3.1 SPH Approximation

For a quantity A evaluated at particle i, the SPH approximation is written as

$$A_i \approx \sum_j m_j \frac{A_j}{\rho_j} W(\|\mathbf{r}_i - \mathbf{r}_j\|, h), \tag{1}$$

where m_j is the particle mass, ρ_j is the density, h is the smoothing length, and W is a compact-support kernel [5, 7]. We use the two-dimensional cubic spline with support radius $2h$,

$$W(q,h) = \frac{5}{14\pi h^2} \begin{cases} (2-q)^3 - 4(1-q)^3, & 0 \le q < 1, \\ (2-q)^3, & 1 \le q < 2, \\ 0, & q \ge 2, \end{cases} \qquad q = \frac{\|\mathbf{r}_i - \mathbf{r}_j\|}{h}. \tag{2}$$

Particles are initialised on a square lattice with spacing Δx and carry the mass $m = \rho_0 \Delta x^2$, so that a particle inside a regular lattice sums to the rest density ρ_0. The smoothing length is set to $h = 1.3\,\Delta x$, which gives about 20 neighbours per particle. Larger ratios make the cubic spline prone to the pairing instability [4], in which particles clump into rows; this is examined in Section 4.6.

3.2 Density and Equation of State

The density is obtained by direct summation over all particles within the kernel support, including the particle itself and the wall particles,

$$\rho_i = m_i W(0,h) + \sum_{j \ne i} m_j W_{ij}. \tag{3}$$

The fluid is treated as weakly compressible [8, 3]. Pressure follows the Tait equation

$$p_i = B\left[\left(\frac{\rho_i}{\rho_0}\right)^{\gamma} - 1\right], \qquad B = \frac{\rho_0 c^2}{\gamma}, \qquad \gamma = 7, \tag{4}$$

where c is the numerical speed of sound; negative pressures are set to zero to avoid tensile clumping at the free surface. The density error scales with the squared Mach number $\mathrm{Ma}^2 = (v/c)^2$, so c is chosen as ten times the largest expected flow speed. Estimating that speed by free fall from the top of the fluid body of height H gives

$$c = 10\sqrt{2gH}, \tag{5}$$

which targets density errors of about one percent.

3.3 Momentum Equation

The acceleration of a fluid particle is the sum of a pressure term, a viscous term and gravity,

$$\mathbf{a}_i^p = -\sum_j m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2}\right) \nabla_i W_{ij}, \tag{6}$$

$$\mathbf{a}_i^\nu = 2\nu \sum_j \frac{m_j}{\rho_j} \frac{\mathbf{v}_{ij} \cdot \mathbf{r}_{ij}}{\|\mathbf{r}_{ij}\|^2 + 0.01h^2} \nabla_i W_{ij}, \tag{7}$$

$$\mathbf{a}_i = \mathbf{a}_i^p + \mathbf{a}_i^\nu + \mathbf{g}, \tag{8}$$

with $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$. The symmetric pressure term conserves linear momentum; the viscous term follows [7, 9] and also damps the acoustic oscillations that a weakly compressible fluid carries.

3.4 Boundary Conditions

Walls are built from fixed particles placed on the same lattice as the fluid. Enough layers are used to fill the full kernel support of a fluid particle touching the wall ($\lceil 2h/\Delta x \rceil = 3$ layers for $h = 1.3\Delta x$), so that wall particles enter the density sum like fluid at rest density.

Wall particles carry no state of their own, so a pressure has to be assigned to them. If the wall term used only p_i/ρ_i^2 [2], the wall would provide half the support that a fluid neighbour provides; the first fluid layer then has to compress until its pressure is roughly doubled, which pushes the second layer away and leaves a visible gap above the floor. Instead the wall pressure is extrapolated from the surrounding fluid following Adami et al. [1],

$$p_b = \frac{\sum_f p_f W_{bf} + \mathbf{g} \cdot \sum_f \rho_f \left(\mathbf{r}_b - \mathbf{r}_f\right) W_{bf}}{\sum_f W_{bf}}, \tag{9}$$

where the sums run over the fluid neighbours f of the wall particle b and the second term adds the hydrostatic pressure difference across the wall. The wall density ρ_b follows from inverting Eq. (4), and the fluid–wall pressure force uses the same symmetric form as the fluid–fluid force.

As a safety net against penetration during violent impacts, a fluid particle that comes closer than $\Delta x/2$ to a wall particle is projected back to that distance and its normal velocity is reflected with a restitution of 0.5. Fluid at rest sits a full spacing away from the wall, so the projection does not act in quiescent regions.

3.5 Time Integration

Positions and velocities are advanced with the semi-implicit (symplectic) Euler scheme, $\mathbf{v}^{n+1} = \mathbf{v}^n + \Delta t\, \mathbf{a}^n$, $\mathbf{r}^{n+1} = \mathbf{r}^n + \Delta t\, \mathbf{v}^{n+1}$. The time step is chosen adaptively in every substep as the most restrictive of the acoustic CFL, force, and viscous conditions [7],

$$\Delta t = \min\left(0.4\,\frac{h}{c + \max_i \|\mathbf{v}_i\|},\; 0.25\sqrt{\frac{h}{\max_i \|\mathbf{a}_i\|}},\; 0.125\,\frac{h^2}{\nu}\right). \tag{10}$$

Each rendered frame advances the simulation by $1/60\,\mathrm{s}$, split into as many substeps as these limits require. For the stiff fluids used here the acoustic condition dominates.

3.6 Neighbour Search

Neighbour lists are rebuilt once per substep by testing all particle pairs against the kernel support radius $2h$. The cost of this search is $\mathcal{O}(N^2)$; it is adequate for the scene sizes used in this work (about 1.6×10^3 particles) but is the main limitation for larger scenes.

3.7 Software Structure

The solver is separated into the following components:

- `Particle` stores particle state and physical properties.
- `NeighborSearch` constructs local particle neighborhoods.
- `SPHSolver` performs the density, wall-pressure, force, and integration steps and chooses the adaptive time step.
- `Scenes` builds the test scenes (tank walls, fluid body, and matching solver parameters).
- `Renderer` displays the scene and exposes runtime controls.

This separation allows the numerical code to be tested and run without an interactive graphics window. A headless program (`sph_experiments`) runs all experiments of Section 4 and writes their data as CSV files, from which the figures are generated.

3.8 Experimental Setup

All quantities are given in screen units (pixels and seconds). Results are reported in dimensionless form ($p/\rho_0 g H$, Z/a, $T = t\sqrt{g/a}$), so the choice of units does not affect them. Two scenes are used (Table 1):

- **Hydrostatic column.** An open tank of width 300 px is filled to $H = 400$ px with fluid at rest. The column has to settle to the hydrostatic pressure $p = \rho_0 g d$ at depth d.
- **Dam break.** A fluid column of width $a = 200$ px and height $H = 400$ px (aspect ratio $n^2 = H/a = 2$) stands in the lower-left corner of a closed tank of width 800 px and collapses under gravity at $t = 0$ [6].

4 Results

All data in this section are produced by the experiment runner `sph_experiments` and plotted by the script `plots/make_plots.py`; rerunning both regenerates every figure.

Table 1: Default parameters of both scenes. Parameter studies vary one quantity at a time.

Quantity	Symbol	Value
Particle spacing	Δx	$10\,\mathrm{px}$
Smoothing length	h	$13\,\mathrm{px}$ $(1.3\,\Delta x)$
Rest density	ρ_0	300
Gravity	g	$200\,\mathrm{px/s^2}$
Speed of sound	c	$10\sqrt{2gH} = 4000\,\mathrm{px/s}$
Viscosity	ν	$150\,\mathrm{px^2/s}$
Fluid / wall particles (column)		1200 / 420
Fluid / wall particles (dam break)		800 / 828
Frame time		$1/60\,\mathrm{s}$

4.1 Correctness Tests

The repository includes CTest unit tests for the neighbour search, for the density summation (an isolated particle carries $m\,W(0,h)$; a particle inside a regular lattice recovers ρ_0 to within 0.01 %), for the slope of the equation of state ($\partial p/\partial\rho = c^2$ at ρ_0), and a short hydrostatic-column regression test that checks the CFL limit, containment, the density error and the hydrostatic pressure profile.

4.2 Hydrostatic Column

Figure 1 compares the particle pressures of the settled column with the hydrostatic solution $p = \rho_0 g d$. Instantaneous pressures scatter around the solution with a mean deviation of about 10 % of $\rho_0 g H$. Averaged over 2 s, the scatter shrinks to 2.4 % and the mean pressure below the surface layer matches the hydrostatic value to within 0.3 %. The instantaneous scatter is therefore not a bias but acoustic noise: pressure waves travel through the column much faster than the frame rate.

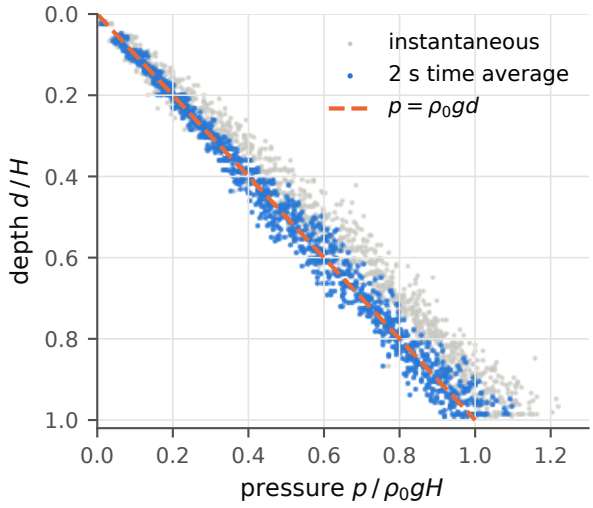


Figure 1: Pressure against depth in the settled hydrostatic column ($t = 8$–$10\,\mathrm{s}$). Grey: instantaneous particle pressures; blue: pressures averaged over 2 s; dashed: hydrostatic solution.

Figure 2 shows how the column settles. The particles start on a lattice at exactly ρ_0 and therefore without pressure, so the column first falls slightly and then oscillates acoustically. Viscosity damps the mean kinetic energy by more than an order of magnitude within 10 s.

The maximum density error settles at 0.56 % (mean 0.25 %), close to the estimate $\Delta\rho/\rho_0 \approx gH/c^2 = 0.5\,\%$ for the compression at the bottom of the column. The adaptive time step settles at $\Delta t \approx 1.2 \times 10^{-3}$ s, i.e. 13 substeps per frame.

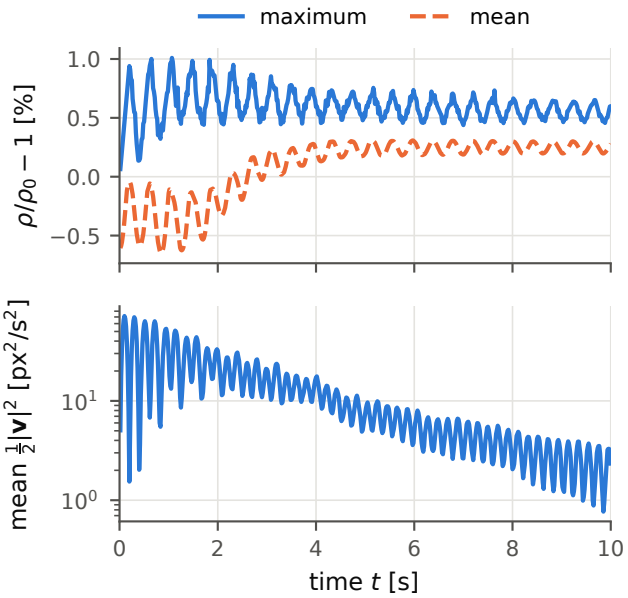


Figure 2: Settling of the hydrostatic column: density error (top) and mean kinetic energy per unit mass (bottom, logarithmic).

4.3 Dam Break

Figure 4 shows the collapse of the water column, coloured by pressure. The front position Z along the floor is shown in Fig. 3 in the dimensionless form of Martin and Moyce [6]. The front reaches $Z/a = 1.87$ at $T = 1$ and 3.62 at $T = 2$, and hits the opposite wall ($Z/a = 4$) at $T = 2.18$. After an initial acceleration phase the front moves at a nearly constant speed. The shallow-water solution of Ritter [10], $Z/a = 1 + 2nT$, assumes that the front starts at full speed and bounds the motion from above.

Until the impact on the right wall the maximum density error stays below 1.5 % and the Mach number below 0.11, as intended by the choice of c. At the impact the flow is deflected upwards and the splash reaches Ma = 0.23, more than twice the free-fall estimate; the density error peaks at 8.9 % for a few frames. The speed of sound chosen from a free-fall estimate is therefore adequate for the collapse itself but marginal for the impact.

4.4 Speed of Sound

The speed of sound trades compressibility against cost (Fig. 5). Doubling c reduces the density error of the hydrostatic column roughly fourfold (10.2 %, 3.0 %, 0.75 % and 0.35 % for $c/\sqrt{2gH} = 2.5, 5, 10, 20$), following the gH/c^2 estimate until noise dominates at the stiffest setting. In the dam break, the time-averaged peak density error decreases only proportionally to $1/c$ (8.1 % to 0.8 %), because it is set by local dynamic compression rather than by the hydrostatic load. The number of substeps grows almost in proportion to c (from 5 to 28 per frame for an eightfold increase) since the acoustic CFL condition dominates. A stiffer fluid is also noisier: the frame-to-frame pressure change grows from 2.3 % to 40 % of $\rho_0 gH$, because the same small density fluctuation is multiplied by $B \propto c^2$. The front position is insensitive to c (Z/a at $T = 1.5$ varies by less than 1 %).

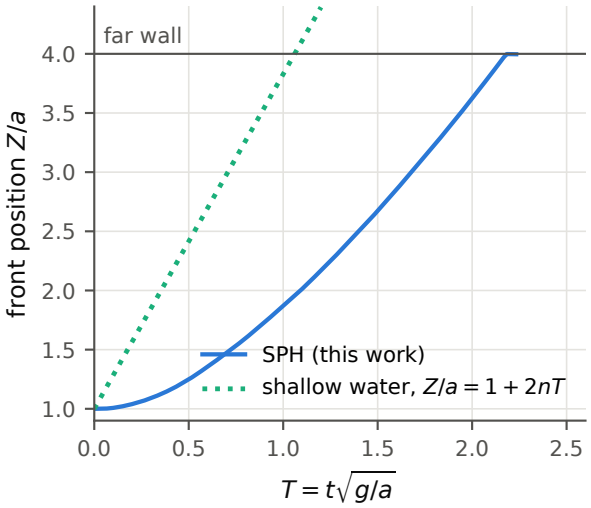


Figure 3: Front position of the dam break in dimensionless form (a: initial column width, $n^2 = H/a = 2$).

4.5 Viscosity

Viscosity mainly controls the acoustic noise (Fig. 6). With $\nu = 50$ the instantaneous pressure of the settled column deviates by 51 % from the hydrostatic value on average; $\nu = 150$ reduces this to 10 %. Larger viscosities do not reduce the instantaneous error further (15–17 %), while the time-averaged error stays between 1.5 and 3.2 % for all values. The flow itself becomes slower with increasing viscosity: at $T = 1.5$ the front has reached $Z/a = 2.74$ for $\nu = 50$ and 2.53 for $\nu = 1000$. The default $\nu = 150$ is a compromise between noise and numerical dissipation.

4.6 Kernel Support and Pairing Instability

Figure 7 shows how the ratio $h/\Delta x$ changes the particle arrangement in the settled column. At $h/\Delta x = 1.2$ the particles remain on the initial lattice, visible as sharp peaks every 10 px; the fluid behaves like a crystal. At 1.3 the particles rearrange into a disordered, liquid-like state. At 1.6 and 2.0 (33 and 50 neighbours instead of about 20) the particles collapse into horizontal rows 18–20 px apart, roughly twice the initial spacing, while the spacing within a row shrinks: the pairing instability of the cubic spline [4]. The rows also make the pressure noisier: at $h/\Delta x = 2.0$ the frame-to-frame pressure change is four times larger than at 1.3. The front position of the dam break is not affected by the ratio.

4.7 Density Fields

Figure 8 compares the density error of the dam break at $T = 2$ for all parameter variations. With $c = 2.5\sqrt{2gH}$ the lower part of the fluid is compressed by several percent; from $c = 10\sqrt{2gH}$ on, the bulk is within about ± 1 % of ρ_0. In all runs the free surface shows a density deficit, because the kernel support of surface particles is only partially filled. Viscosity and $h/\Delta x$ hardly change the density field at this resolution.

4.8 Discussion

The weakly compressible model reproduces the hydrostatic pressure with a time-averaged accuracy of a few percent and keeps density errors at about one percent for the chosen speed of

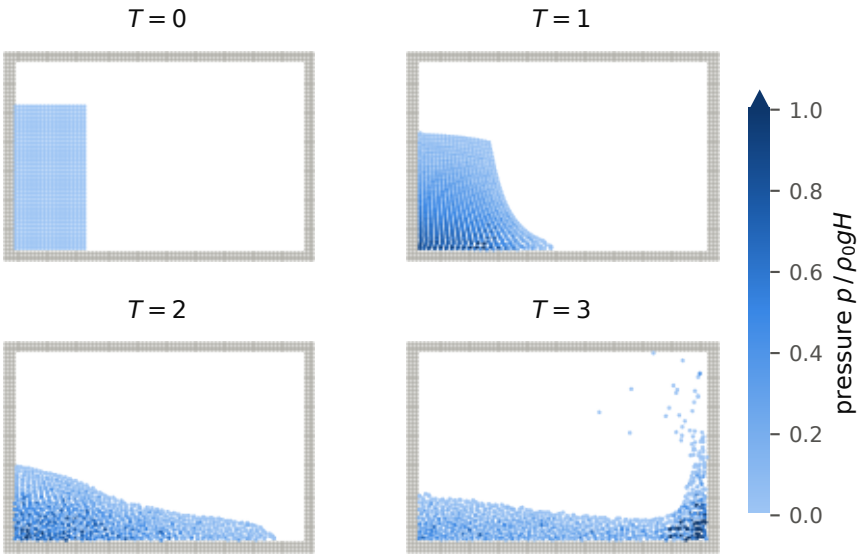


Figure 4: Dam break at $T = t\sqrt{g/a} = 0, 1, 2, 3$. Fluid particles are coloured by pressure, smoothed over 0.05 s per particle to suppress acoustic noise; wall particles are grey.

sound. Its main weakness is acoustic pressure noise: instantaneous particle pressures scatter by about 10 % of the hydrostatic pressure even in a fluid at rest. The interactive renderer therefore colours particles by a per-particle moving average of the pressure. Methods that diffuse density perturbations (δ-SPH) or solve for pressure implicitly (IISPH, DFSPH) would remove this noise at the source.

Further limitations are the $\mathcal{O}(N^2)$ neighbour search, which limits the resolution to about 10^3 fluid particles at interactive rates; the projection safety net at the walls, which is not part of the SPH formulation; and the free-fall estimate of the flow speed used to choose c, which underestimates the speeds reached at the wall impact.

5 Conclusion

This project provides a small and inspectable two-dimensional SPH solver with a weakly compressible equation of state, adaptive time stepping, extrapolated wall pressures, an interactive visualisation, and a reproducible experiment pipeline.

With the speed of sound set to ten times the free-fall velocity, the settled hydrostatic column keeps its density within about 0.6 % of the rest density and reproduces the hydrostatic pressure to within a few percent when averaged over time. The dam break stays below 1.5 % density error until the water hits the opposite wall. The parameter studies show that the speed of sound trades compressibility against cost and pressure noise, that viscosity is needed to damp acoustic oscillations but slows the flow, and that the smoothing length must stay close to 1.3 particle spacings: larger values trigger the pairing instability, smaller ones freeze the particles on their initial lattice.

Future work could replace the all-pairs neighbour search by a uniform grid to reach higher resolutions, reduce the pressure noise with density diffusion (δ-SPH) or an implicit pressure solver (IISPH, DFSPH), and compare the dam-break front quantitatively with the measurements of Martin and Moyce.

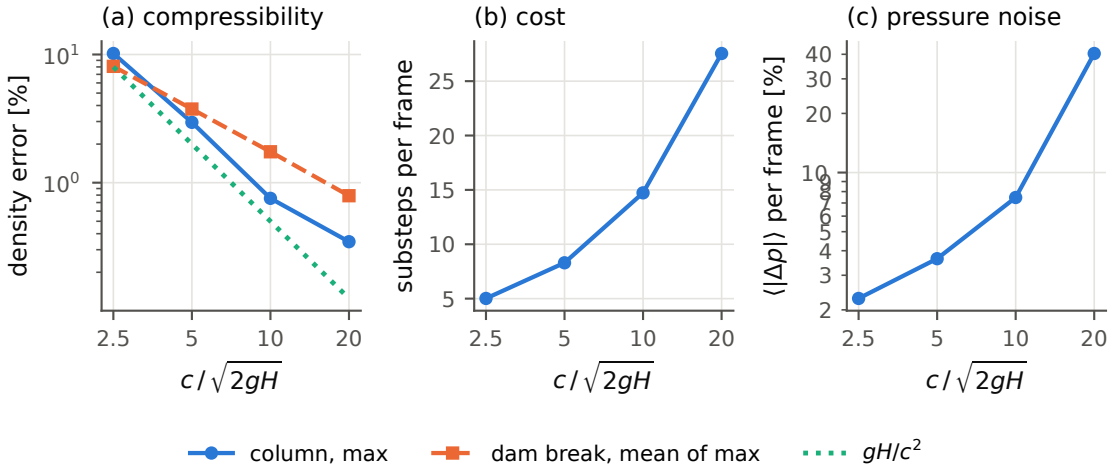


Figure 5: Influence of the speed of sound: (a) density error with the hydrostatic estimate gH/c^2, (b) substeps per 1/60 s frame in the dam break, (c) mean frame-to-frame pressure change in the settled column.

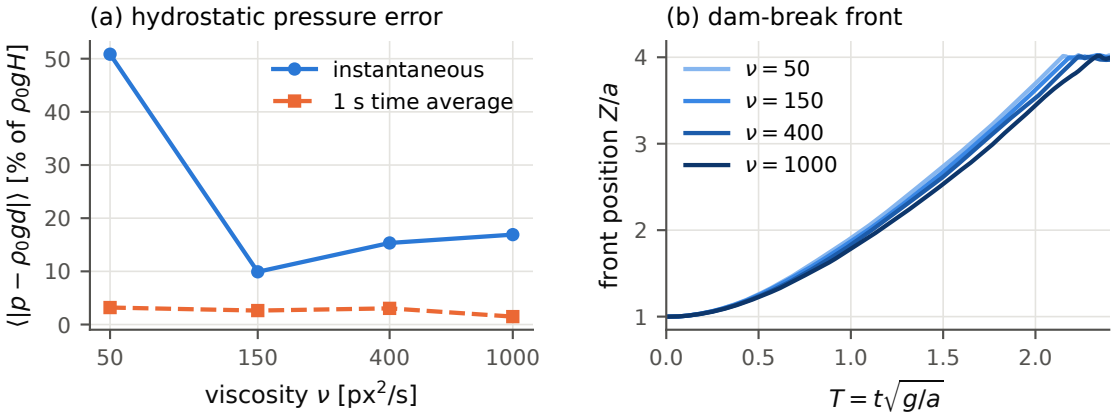


Figure 6: Influence of the viscosity ν: (a) mean pressure error of the settled column, instantaneous and averaged over 1 s; (b) front position of the dam break.

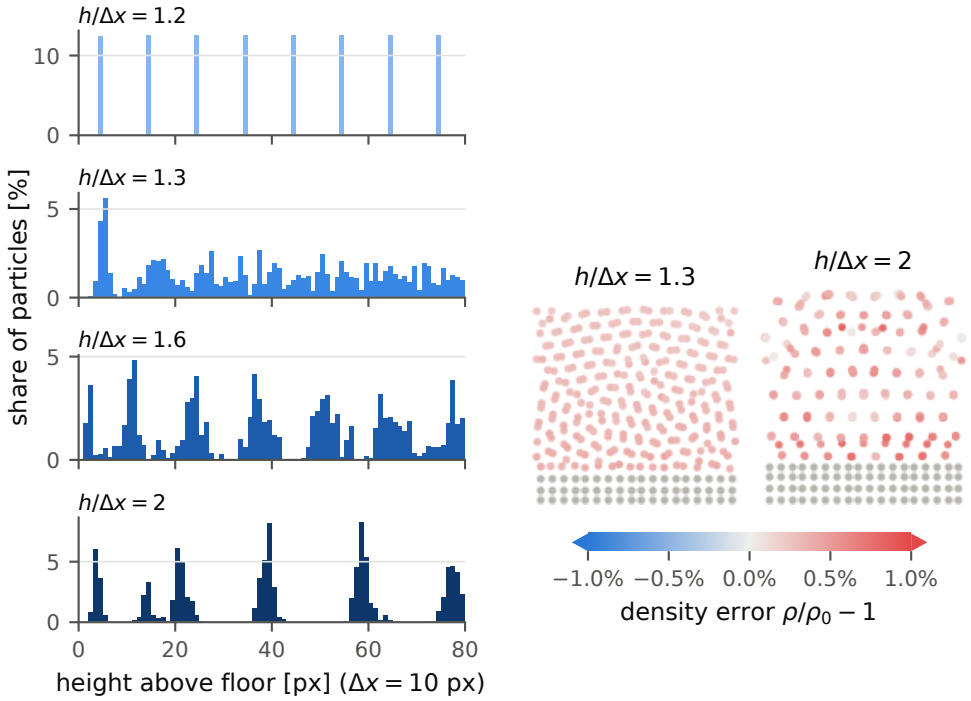


Figure 7: Particle arrangement in the settled hydrostatic column for different $h/\Delta x$. Left: distribution of particle heights above the floor (1 px bins, central region). Right: the bottom of the column for $h/\Delta x = 1.3$ and 2.0, coloured by density error.

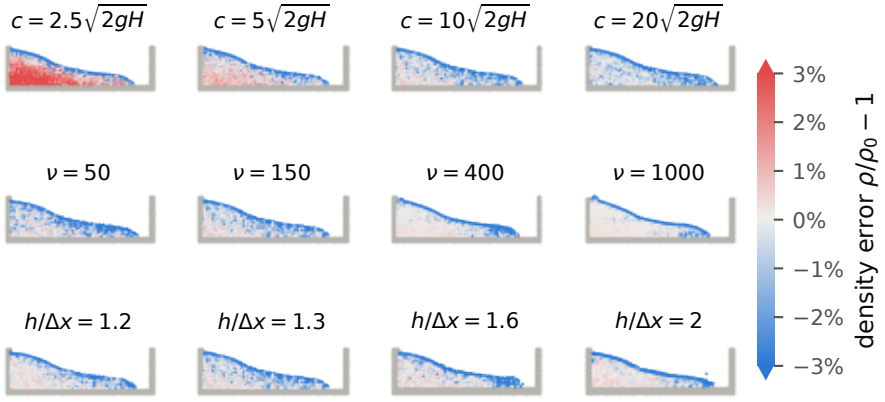


Figure 8: Density error $\rho/\rho_0 - 1$ in the dam break at $T = 2$ (lower part of the tank). Rows vary the speed of sound, the viscosity, and the ratio $h/\Delta x$; the default setup is $c = 10\sqrt{2gH}$, $\nu = 150$, $h/\Delta x = 1.3$.

References

[1] Stefan Adami, Xiangyu Y. Hu, and Nikolaus A. Adams. A generalized wall boundary condition for smoothed particle hydrodynamics. *Journal of Computational Physics*, 231(21):7057–7075, 2012.

[2] Nadir Akinci, Markus Ihmsen, Gizem Akinci, Barbara Solenthaler, and Matthias Teschner. Versatile rigid-fluid coupling for incompressible SPH. *ACM Transactions on Graphics*, 31(4):62:1–62:8, 2012.

[3] Markus Becker and Matthias Teschner. Weakly compressible SPH for free surface flows. In *Proceedings of the 2007 ACM SIGGRAPH/Eurographics Symposium on Computer Animation*, pages 209–217, 2007.

[4] Walter Dehnen and Hossam Aly. Improving convergence in smoothed particle hydrodynamics simulations without pairing instability. *Monthly Notices of the Royal Astronomical Society*, 425(2):1068–1082, 2012.

[5] Leon B. Lucy. A numerical approach to the testing of the fission hypothesis. *The Astronomical Journal*, 82:1013–1024, 1977.

[6] J. C. Martin and W. J. Moyce. Part IV. an experimental study of the collapse of liquid columns on a rigid horizontal plane. *Philosophical Transactions of the Royal Society of London. Series A*, 244(882):312–324, 1952.

[7] Joseph J. Monaghan. Smoothed particle hydrodynamics. *Annual Review of Astronomy and Astrophysics*, 30:543–574, 1992.

[8] Joseph J. Monaghan. Simulating free surface flows with SPH. *Journal of Computational Physics*, 110(2):399–406, 1994.

[9] Joseph P. Morris, Patrick J. Fox, and Yi Zhu. Modeling low reynolds number incompressible flows using sph. *Journal of Computational Physics*, 136(1):214–226, 1997.

[10] A. Ritter. Die Fortpflanzung der Wasserwellen. *Zeitschrift des Vereines Deutscher Ingenieure*, 36:947–954, 1892.